

HDOC - Day 32

Matrix Commutative Properties

Algorithm

Use separate functions. For addition, calculate A+B and B+A and compare them. For multiplication, use square matrices so both AB and BA exist, calculate both products, display them and use a test case where they differ to demonstrate that matrix multiplication is not commutative in general.

Test Case Table

T C	Input/Operation	Expected	Actual	Result
1	2x2 matrices, addition	A+B=B+A	Equal	Pass
2	2x2 matrices, multiplication	AB != BA	Not equal	Pass

C Program

```
#include <stdio.h>
void readM(int n,int a[n][n]){int i,j;for(i=0;i<n;i++)for(j=0;j<n;j++)scanf("%d",&a[i][j]);}
void show(int n,int a[n][n]){int i,j;for(i=0;i<n;i++){for(j=0;j<n;j++)printf("%d ",a[i][j]);printf("\n");}}
int same(int n,int a[n][n],int b[n][n]){int i,j;for(i=0;i<n;i++)for(j=0;j<n;j++)if(a[i][j]!=b[i][j])return 0;return 1;}
void add(void){int n,i,j;printf("Enter order: ");scanf("%d",&n);int A[n][n],B[n][n],X[n][n],Y[n][n];printf("A: ");readM(n,A);printf("B: ");readM(n,B);
for(i=0;i<n;i++)for(j=0;j<n;j++){X[i][j]=A[i][j]+B[i][j];Y[i][j]=B[i][j]+A[i][j];}printf("A+B:\n");show(n,X);printf("B+A:\n");show(n,Y);printf(same(n,X,Y)?"Addition is co
void mult(void){int n,i,j,k;printf("Enter order: ");scanf("%d",&n);int A[n][n],B[n][n],X[n][n],Y[n][n];printf("A: ");readM(n,A);printf("B: ");readM(n,B);
for(i=0;i<n;i++)for(j=0;j<n;j++){X[i][j]=Y[i][j]=0;for(k=0;k<n;k++){X[i][j]+=A[i][k]*B[k][j];Y[i][j]+=B[i][k]*A[k][j];}}
printf("A*B:\n");show(n,X);printf("B*A:\n");show(n,Y);printf(!same(n,X,Y)?"Multiplication is not commutative.\n":"A*B equals B*A for this input.\n");}
int main(){int ch;do{printf("\n1.Addition commutative\n2.Multiplication not commutative\n3.Exit\nChoice: ");scanf("%d",&ch);switch(ch){case 1:add();break;case 2:mult();br
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day32.c -o day32
puppie@Anvikas-MacBook-Air Desktop % ./day32
Choice: 1 Order: 2
A: 1 2 3 4 B: 5 6 7 8
A+B: 6 8 / 10 12
B+A: 6 8 / 10 12
Addition is commutative.
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day32
Choice: 2 Order: 2
A: 1 2 0 1 B: 1 0 3 1
A*B: 7 2 / 3 1
B*A: 1 2 / 3 7
Multiplication is not commutative.
```

Result: The program was compiled, executed and tested successfully.